

Software Requirements Specification – Healthcare resources Optimization System

" Medical Equipment Management"

Team Members:

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Chapter1:

Business Requirements:

Device Tracking: The system should allow doctors to easily locate medical devices within the hospital using sensors.

Device Monitoring: The system must provide doctors with real-time updates on the Location of medical devices.

Efficiency Improvement: The system should help improve hospital operations by reducing the time needed to find and assess medical equipment.

User Accessibility: Doctors should be able to access the system quickly and easily to get location and information from any hospital device.

Software Stockholders:

- Head medical services
- Head medical operations
- Doctors
- Nurses

Software Type: Interactive distributed

Chapter2 (Software Requirements Specification phase):

Requirements Gathering:

Interview: We conducted a real interview with a doctor and a patient. Based on this, we gathered a wealth of useful information and requirements. We learned that they frequently face shortages of medical equipment and resources, which forces them to seek alternative solutions to manage their medical needs effectively. This shortage significantly impacts their ability to provide timely and efficient care, leading to delays and potential compromises in patient care.

Our software specifically addresses this issue by focusing on ensuring the constant availability of medical equipment and resources. It includes features designed to monitor inventory levels in realtime, automatically alerting staff when supplies are running low. Additionally, the software tracks the locations of all medical equipment, helping to quickly identify and retrieve needed items. This feature reduces the time spent searching for equipment and minimizes downtime caused by resource shortages.

By improving the management of medical resources and equipment, our software aims to enhance operational efficiency and support better patient care. This proactive approach ensures that medical staff have the tools they need when they need them, thereby improving overall healthcare delivery.

Interview Q&A:

- Q1: What problems do you face with medical equipment in your practice?

A1: We often run out of essential medical equipment and supplies. This forces us to look for alternative solutions and causes delays in patient care.

- Q2: How do these equipment shortages affect your daily work?

A2: Shortages cause delays in treating patients and sometimes make us reschedule appointments. This affects the quality of care and adds stress to both staff and patients.

- Q3: Which types of medical equipment are most often missing?

A3: We frequently lack important equipment like dialysis machines, blood pressure monitors, and other tools needed for kidney care.

- Q4: How does your current system manage medical supplies? Are there any problems with it?

A4: The current system doesn't track supplies well and doesn't give timely updates. This leads to running out of essential equipment unexpectedly.

- Q5: What problems do you have with the notifications for low stock levels?

A5: The notifications are often unreliable and come too late, so we sometimes run out of supplies before we know we need more.

- Q6: How do system crashes or updates affect your work?

A6: System crashes and disruptive updates slow down our work and cause delays. This also adds extra stress and reduces our efficiency.

- Q7: What changes would help manage medical equipment and supplies better?
A7: We need a system that tracks supplies in real-time, alerts us when stock is low, and is easy to use. It should also have minimal disruptions.
- Q8: Are there any features missing from the current system that you need?
A8: Yes, we need features like real-time tracking of equipment locations and automatic alerts when supplies are low. These are important for keeping things running smoothly.
- Q9: How important is it to have real-time tracking of equipment and supplies?
A9: Real-time tracking is very important. It helps us find equipment quickly, manage supplies better, and avoid delays in patient care.
- Q10: What would be the best solution to fix the equipment shortages and improve your practice?
A10: The best solution would be a software system that tracks supplies in real-time, sends alerts for low stock, and helps us find equipment easily. This would improve efficiency and ensure we have what we need for patient care.

Name	Position	Purpose of interview	Meeting
Healthcare Professional Interview	Nephrologist.	To collect information about the software.	Thursday, 12 September, 7 PM to 8:30 PM.

Functional requirements:

Stockholders

1. The user should be able to create an account
 - 1.1 The system shall take the name and the password from the user.
 - 1.2 The system shall verify the name if it exists in the database a message error will appear (the name exists).
 - 1.3 If the name doesn't exist in the database the system shall store the name and the password in the database.

Stockholders

2. The user should be able to log in.
 - 2.1 The system shall take the name and the password from the user.
 - 2.2 The system shall check if the name exists in the database.
 - 2.3 If the name doesn't exist in the database an error message will appear (The name is wrong).
 - 2.4 The system shall check if the password exists in the database.
 - 2.5 If the password doesn't exist in the database an error message will appear (The password is wrong).
 - 2.6 If the name and the password exist in the database login.

Stockholders

3. The user should be able to verify their account.
 - 3.1 After account creation, the system shall send a verification email or message to the user.
 - 3.2 The user shall receive a verification link or code.
 - 3.3 The system shall check the verification link or code when the user attempts to verify their account.
 - 3.4 If the link or code is valid, the system shall activate the user's account.
 - 3.5 If the link or code is invalid, an error message shall appear stating, "Verification failed."

Head medical services

4. The user shall be able to add new equipment to the database.
 - 4.1 The system shall be able to take the name and the ID from the user.
 - 4.2 The system shall be able to take the section's name to insert the equipment inside.
 - 4.3 The system shall be able to take the equipment job from the user.
 - 4.4 The system shall be able to check if the equipment exists in the database. 4.5 If the system finds the equipment exists a message error will appear (the equipment exists).
 - 4.6 If the system doesn't find the equipment, then the equipment will be added to the database.

Head medical services

- 5. The user shall be able to delete equipment.
- 5.1 The system shall be able to take the name or ID of the equipment.
- 5.2 If the equipment is there, the user can delete the equipment.
- 5.3 If the system can't find the equipment a message error will appear (sorry can't find the equipment).

Head medical services

- 6. The user shall be able to edit equipment.
- 6.1 The system shall be able to take the name or the ID for the equipment.
- 6.2 If the equipment is found, the user can change its details (like name, ID, etc).
- 6.3 If the system can find the equipment a message will appear (The equipment has been successfully updated).
- 6.4 If the system can't find the equipment a message error will appear (sorry can't find the equipment).

Stockholders

- 7. The user should be able to search for certain equipment.
- 7.1 The system shall be able to take the name or the ID for the equipment.
- 7.2 If the system finds the equipment the information will appear.
- 7.3 If the system can't find the equipment a message error will appear (sorry can't find the equipment).

Doctor

- 8. The user shall be able to book an appointment for medical equipment.
- 8.1 The system shall allow the user to search for the patient's data.
- 8.2 The system shall enable the user to select the required equipment based on the patient's needs.
- 8.3 The system shall reveal the available equipment and its status.
- 8.4 The system shall allow the user to reserve the equipment for a future time when it becomes available.
- 8.5 The system shall send a confirmation message to the user indicating that the equipment has been successfully reserved.
- 8.6 If the equipment is not available (it does not exist in the inventory) The system shall send a notification to the Head of Medical Services informing them that the equipment is unavailable for booking.
- 8.7 The system shall reveal an error message to the user stating, "Sorry, the equipment is not available."

Sensors location

- 9. The user should be able to get information about the real-time location of medical equipment.
- 9.1 The system should take the name or ID of the equipment.
- 9.2 The system should find a specific sensor that is linked to the equipment using the equipment's name or ID.

9.3 The system should get the equipment's location coordinates from the sensor once it's found.

9.4 The system should send the location coordinates back to the user.

Head medical operations

10. The user shall be able to manage maintenance schedules for medical equipment.

10.1 The system shall allow users to schedule maintenance activities, including setting dates, times, and types of maintenance.

10.2 The system shall send notifications or reminders to users about upcoming scheduled maintenance.

Stockholders

11. The user should be able to log out.

11.1 The system shall take the user's request to log out.

11.2 The system shall ask the user to confirm their intention to log out.

11.3 If the user confirms, the system shall end the user session.

11.4 If the user cancels, the system shall keep the user logged in.

11.5 After logging out, the system shall clear all user information.

Non-Functional requirements:

1. Availability

The software shall be available 24/7.

2. Failure

The system must not be down for more than 4 min. In its 10-year life-span.

3. Response Time

The situation of the medical equipment must appear within one second.

4. Throughput

The system shall serve more than 10,000 users per day.

5. Security

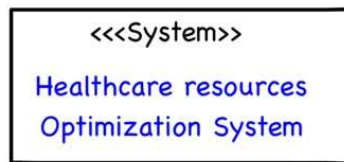
Data can only be accessed and modified by the stockholders.

6. Usability

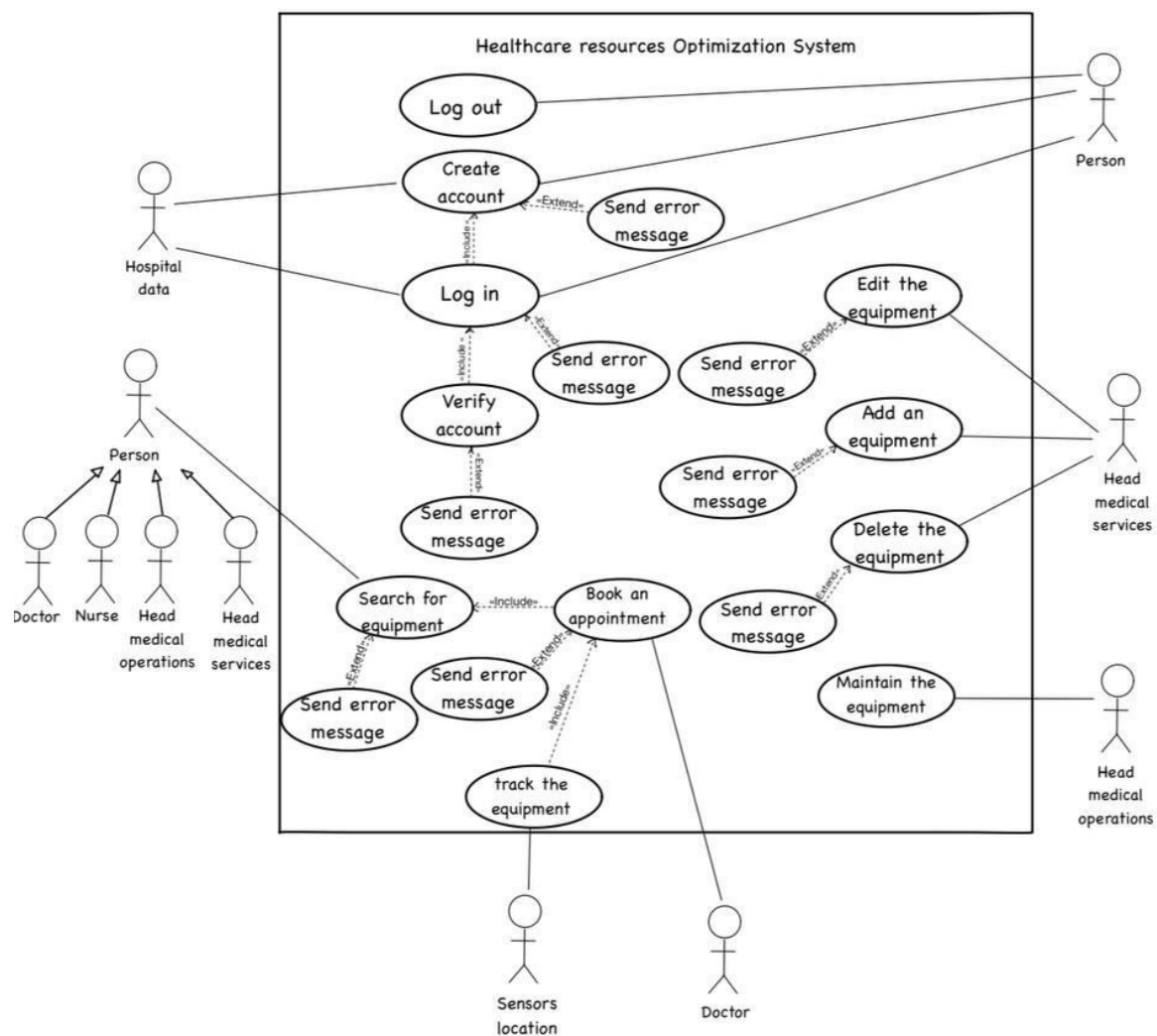
The system shall have a help menu that will explain all functions and how to achieve common tasks.

Chapter3: Software Modeling

■ Context Diagram



■ Use case Diagram



▪ Use case Scenario

1. create account

Field	Description
Actors	Person
Description	A person creates a new account in the healthcare resources optimization system.
Data	User personal information (name, email, etc.)
Stimulus	User initiates the "Create account" action
Response	Confirmation that the account has been created successfully
Comments	Appropriate validations are needed for security and user data accuracy

2. Log in

Field	Description
Actors	Person
Description	The user logs into the healthcare resources optimization system with valid credentials.
Data	Username and password
Stimulus	User inputs login details
Response	Access granted to the system upon successful login
Comments	Requires authentication via username and password, with potential multi-factor authentication options

3. Log out

Field	Description
Actors	Person
Description	The user logs out from the healthcare resources optimization system.
Data	None
Stimulus	User clicks the "Log out" button
Response	Session is terminated and user is logged out
Comments	Ensure proper session management to prevent unauthorized access after logout

4. Verify account

Field	Description
Actors	Person
Description	The system verifies the user's account through a confirmation process (e.g., email verification).
Data	Verification code, email, or other account credentials
Stimulus	User initiates account verification
Response	Confirmation that the account is verified
Comments	This step may involve a confirmation email or a mobile OTP code

5. Search for equipment

Field	Description
Actors	Person
Description	The user searches for specific medical equipment in the system.
Data	Equipment ID or keywords
Stimulus	User inputs equipment details or keyword to search
Response	List of available equipment matching the search criteria
Comments	System must have a well-maintained database of equipment with real-time status updates

6. Book an appointment

Field	Description
Actors	Doctor
Description	The user books an appointment with a specific doctor or service using the system.
Data	Patient details, date, and time
Stimulus	User selects a time slot and doctor for the appointment
Response	Confirmation of appointment booking
Comments	System should prevent double-booking and handle appointment conflicts

7. Track the equipment

Field	Description
Actors	Sensors location
Description	The system allows tracking of equipment locations using sensors.
Data	Equipment location data from sensors
Stimulus	User requests the current location of specific equipment
Response	Real-time location of the equipment is provided
Comments	Requires integration with sensor tracking technology to update locations in real time

8. Edit the equipment

Field	Description
Actors	Head medical services
Description	The user edits equipment details in the system, such as updating status or specifications.
Data	Equipment details (name, ID, condition, etc.)
Stimulus	User inputs new details for the equipment
Response	Confirmation that the equipment information has been updated
Comments	Ensure that only authorized personnel can edit critical information

9. Delete the equipment

Field	Description
Actors	Head medical services
Description	The user deletes equipment from the system when it is no longer in use.
Data	Equipment ID or relevant identifying information
Stimulus	User initiates the "Delete equipment" action
Response	Confirmation that the equipment has been deleted from the system
Comments	Deletion should be restricted to prevent accidental or unauthorized removal

10. Maintain the equipment

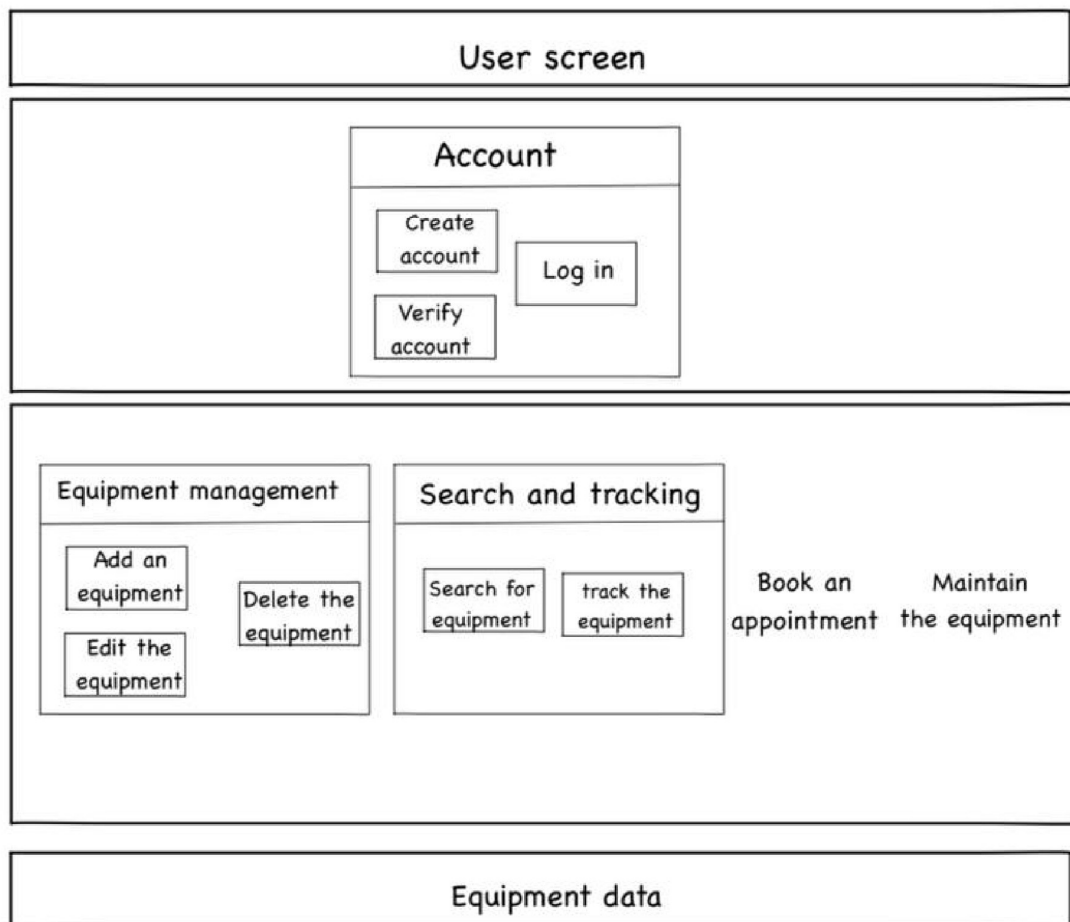
Field	Description
Actors	Head medical operations
Description	The system is updated to reflect maintenance actions on medical equipment.
Data	Maintenance logs, equipment condition
Stimulus	User records or schedules maintenance
Response	Confirmation that maintenance information has been updated
Comments	Regular updates are needed to ensure equipment functionality and safety

11. Add new equipment

Field	Description
Actors	Head medical services
Description	Head of medical services adds new equipment into the system database.
Data	Equipment details (name, ID, section, job).
Stimulus	User initiates the "Add equipment" action
Response	System checks if the equipment exists and adds it if it doesn't exist.
Comments	Appropriate error handling is required if equipment already exists.

Chapter 4 : Software Architectural diagrams

- Layered architectural diagram



▪ Client-Server & Repository architecture

